

Class 11 Protein Structure Prediction with AlphaFold

Rowen Lin (PID: A18363990)

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```
library(bio3d)
```

```
unzip("hipvrdimer_94b5b.result.zip", exdir = "hivprdimer")
```

```
results_dir <- "hivprdimer/hivprdimer_94b5b/"
```

Make a vector of input PDB file names that we can read into R

```
pdb_files <- list.files(path = results_dir, pattern = "*.pdb", full.names = TRUE)
pdb_files
```

```
[1] "hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4"
[2] "hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5"
[3] "hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3"
[4] "hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1"
[5] "hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2"
```

Custom Analysis of AlphaFold Models

```
pdbs <- pdbaln(pdb_files, fit = TRUE, exefile = "msa")
```

Reading PDB files:

```
hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed.
hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed.
hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed.
hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed.
hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed.
.....
```

Extracting sequences

```
pdb/seq: 1   name: hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed.
pdb/seq: 2   name: hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed.
pdb/seq: 3   name: hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed.
pdb/seq: 4   name: hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed.
pdb/seq: 5   name: hivprdimer/hivprdimer_94b5b//hivprdimer_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed.
```

```
pdbs
```

```

1           .           .           .           .           50
[Truncated_Name:1]hivprdimer PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
[Truncated_Name:2]hivprdimer PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
[Truncated_Name:3]hivprdimer PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
[Truncated_Name:4]hivprdimer PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
[Truncated_Name:5]hivprdimer PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKPMIGGI
*****
1           .           .           .           .           50

51          .           .           .           .           99
[Truncated_Name:1]hivprdimer GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]hivprdimer GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]hivprdimer GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]hivprdimer GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]hivprdimer GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
*****
51          .           .           .           .           99
```

Call:

```
pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")
```

Class:

pdbs, fasta

Alignment dimensions:

5 sequence rows; 99 position columns (99 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call

Here we read in all 5 AlphaFold model PDB files and superposed them onto each other using the `pdbaln()` function from `bio3d`.

```
rd <- rmsd(pdbs)
```

Warning in `rmsd(pdbs)`: No indices provided, using the 99 non NA positions

```
print(rd, digits=2)
```

```
hivprdimer_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000
hivprdimer_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000
hivprdimer_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000
hivprdimer_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000
hivprdimer_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000
hivprdimer_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000
hivprdimer_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000
hivprdimer_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000
hivprdimer_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000
hivprdimer_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000
hivprdimer_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000
hivprdimer_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000
hivprdimer_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000
hivprdimer_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000
hivprdimer_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000
hivprdimer_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000
hivprdimer_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000
hivprdimer_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000
hivprdimer_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000
```

```

hivprdimer_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000
hivprdimer_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000
hivprdimer_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000
hivprdimer_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000
hivprdimer_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000
hivprdimer_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000

```

```

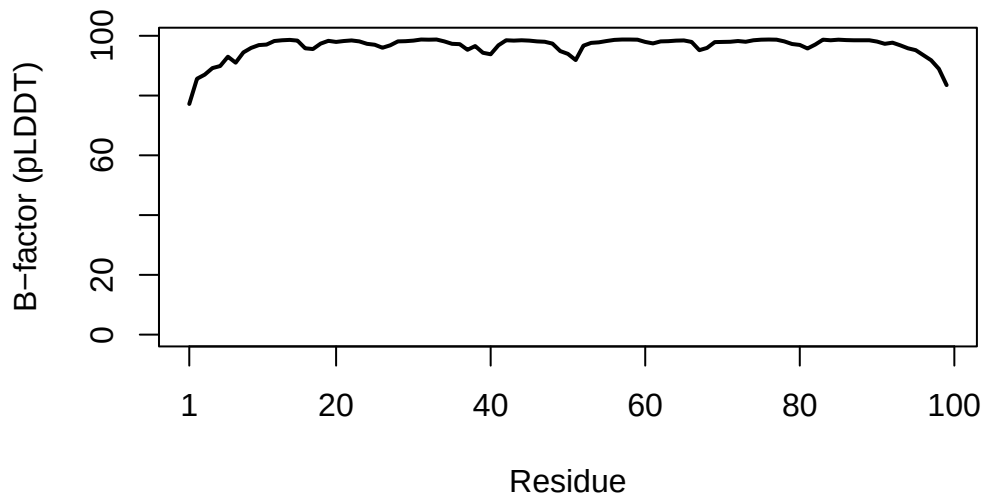
pdb <- read.pdb(pdb_files[1])

plotb3(pdb$atom$b[pdb$calpha], typ="l", lwd=2, sse=pdb,
       ylab="B-factor (pLDDT)")

```

Warning in pdb2sse(sse): No helix and sheet defined in input 'sse' PDB object:
try using dssp()

Warning in plotb3(pdb\$atom\$b[pdb\$calpha], typ = "l", lwd = 2, sse = pdb, :
Length of input 'sse' does not equal the length of input 'x'; Ignoring 'sse'



The plot shows the pLDDT confidence scores per residue for the top ranked model. Values above 70 are considered high confidence. The HIV-Pr monomer shows consistently high pLDDT scores across the entire chain, indicating a reliable prediction throughout.

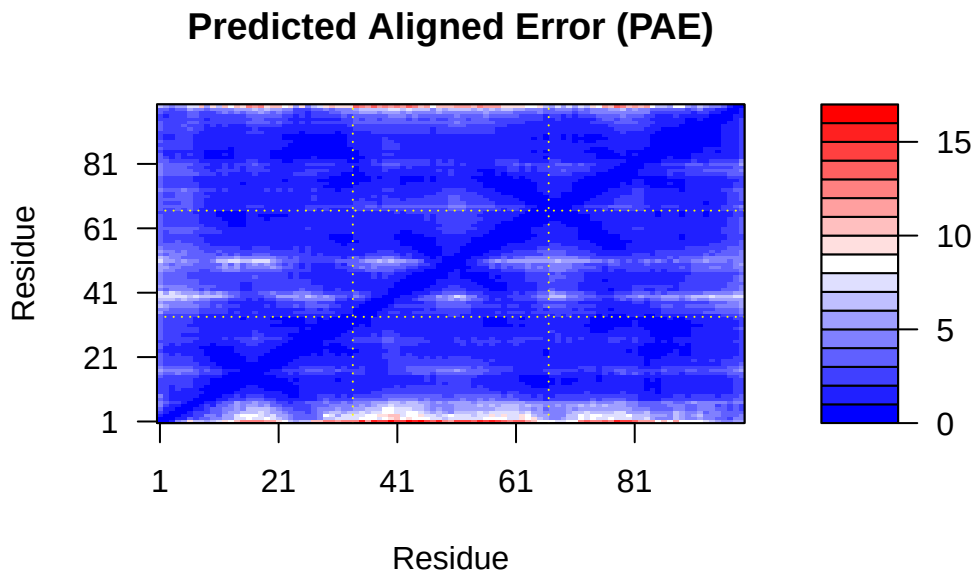
Predicted Alignment Error for domains

Next we examine the Predicted Aligned Error (PAE) matrix, which tells us about the confidence in the relative positions of residue pairs across the structure.

```
library(jsonlite)

pae_file <- list.files(results_dir, pattern="predicted_aligned_error_v1.json",
                        full.names=TRUE)
pae <- fromJSON(pae_file)

plot.dmat(pae$predicted_aligned_error, xlab="Residue", ylab="Residue",
          main="Predicted Aligned Error (PAE)")
```



The PAE matrix shows mostly blue across the entire residue matrix, which shows that AlphaFold is confident in the relative positions of all residue pairs.

Residue conservation from alignment file

```
aln_file <- list.files(results_dir, pattern=".a3m", full.names=TRUE)
aln <- read.fasta(aln_file[1], to.upper=TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

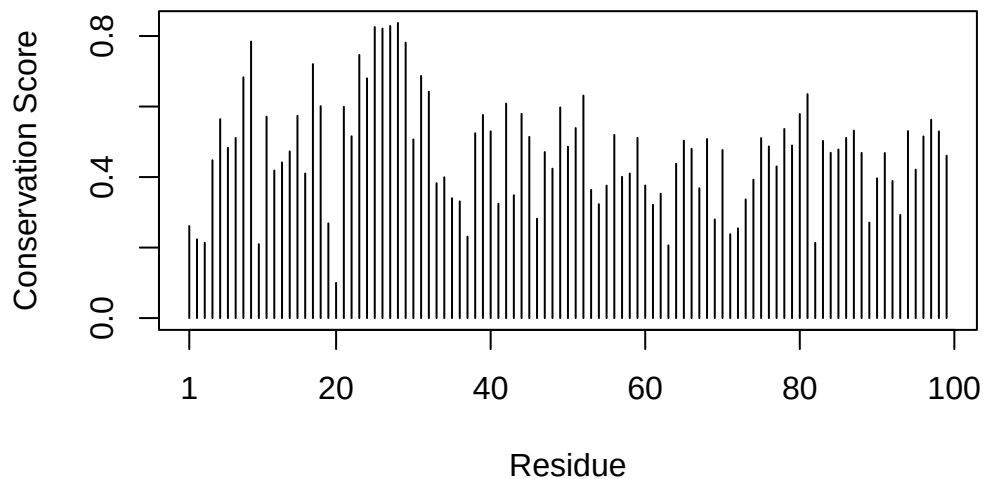
```
[1] 5397 132
```

The alignment has 5397 sequences across 132 positions.

```
sim <- conserv(aln)
plotb3(sim[1:99], sse=pdb, ylab="Conservation Score")
```

Warning in pdb2sse(sse): No helix and sheet defined in input 'sse' PDB object:
try using dssp()

Warning in plotb3(sim[1:99], sse = pdb, ylab = "Conservation Score"): Length of
input 'sse' does not equal the length of input 'x'; Ignoring 'sse'



```
con <- consensus(aln, cutoff=0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
pdb$atom$b[pdb$calpha] <- sim[1:99]
write.pdb(pdb, file="m1_conserv.pdb")
```

Section 10: Summary

In this lab I used ColabFold to generate AlphaFold2 structure predictions for HIV protease. I analyzed the results using bio3d in R, and then I found pLDDT confidence scores, RMSD between models, predicted aligned error, and residue conservation. All 5 of these models showed high confidence and were very similar structures, with the catalytic Asp25 identified as the most conserved residue.